Aggregate biocapacity does not identify its composition: a two-land model of ecological-capital drawdown

Amin Abaee[a,*]

[a]*Independent Researcher, Tehran, Iran*

Abstract

Aggregate biocapacity is a value-weighted total over heterogeneous accounts; the total alone does not identify their composition. We formulate a two-land delay-differential model in which account shortfall triggers one-way conversion from ecological-capital to fast-provisioning area, while the capital-quality state q evolves separately. The conversion contribution to aggregate biocapacity is positive when fast-land yield exceeds the effective per-hectare value of capital land. In a reduced frozen-demand equilibrium relation with $q = q_{\max}$, debt and technology wave off, and restoration disabled, the balance point is unique and the floor collision has no fold or critical slowing down. This branch result does not eliminate trajectory dependence: conversion without restoration leaves realized A_c below its pre-overshoot level after conversion stops, but does not imply perpetual decline at equilibrium. In a representative conditional run, A_c falls 46.2%, $\Upsilon(q)$ falls 10.3%, and the modeled capital contribution $\Upsilon(q)A_c$ falls 51.7% while aggregate B rises 39%; these are model outputs, not empirical stock estimates. In Edition 2026 of the National Footprint and Biocapacity Accounts, world biocapacity rises 24.9% from 1961 to 2023 while B/P falls 52.7% and the global production footprint per person rises 16.7%. Cropland contributes +24.06 percentage points to the total change, Forest biocapacity −5.95, and built-up land +7.44 as an accounting class, not ecological capital. NFA component values are accounting signals, not measurements of biomass, soil condition, wetland extent, fish biomass, or A_c. No comparable global physical-stock inventory is assembled; the proposed scoreboard remains an unpopulated, nonadditive vector pending defensible common boundaries, valuation, normalization, and substitution rules.

Keywords: biocapacity, ecological footprint, land-use conversion, aggregation, identifiability, critical slowing down, delay differential equations, carrying capacity

*Corresponding author. ORCID: 0000-0002-0019-1842.
Email address: amin_abaee@ut.ac.ir (Amin Abaee)

1. Introduction

Carrying capacity is one of ecology's most enduring and most elusive ideas. In classical population biology it is a fixed ceiling; imported into global human ecology it is usually treated as a static boundary, the maximum number of people the Earth can support (cf. Cohen, 1995). Treating the ceiling as fixed overlooks a feedback that matters: a population that overshoots its resource base degrades the base and lowers the future ceiling. This paper is concerned with a subtler failure than a fixed ceiling. It concerns the *measurement* on which monitoring and policy depend.

The standard aggregate measure of annual regenerative supply is biocapacity, expressed in the National Footprint and Biocapacity Accounts (NFA) in global hectares (gha) for each annual account. Under the Global Footprint Network (GFN) convention, annual Ecological Footprint and biocapacity values are reported in gha, not gha yr^{-1} (Global Footprint Network, 2009, 2019). A global hectare is a productivity-normalized hectare whose value incorporates yield and equivalence factors (Borucke et al., 2013; Lin et al., 2018). Because the weighting sits *inside* the unit, an aggregate biocapacity series does not tell a monitor whether capacity rose because land became more productive, because more land was brought into provision, or because ecological capital was converted into provisioning land. The composite-indicator literature warns that additive aggregation turns weights into trade-offs, so that a deficit in one dimension can be offset by a surplus in another (Nardo et al., 2008; Becker et al., 2017; Fischer et al., 2022); biocapacity is exactly such a composite, and the footprint framework has been criticized on related grounds (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; van den Bergh and Grazi, 2015; Galli et al., 2016). The aggregate remains useful as a headline, but only if its components are reported alongside it. What has been missing is a dynamical account of *how* a rising aggregate and a falling capital base can coexist, for how long, and what would be needed to tell them apart.

Coupled human–environment models have largely treated the environment as a single stock or a single aggregate balance: the World3 system (Meadows et al., 1972, 1974, 2004), the Ricardo–Malthus models of Easter Island (Brander and Taylor, 1998; Anderies, 2000), and more recent bidirectionally coupled Earth-system frameworks (Motesharrei et al., 2016; Donges et al., 2017; Calvin and Bond-Lamberty, 2018; Henderson and Loreau, 2018). Estimates of Earth's carrying capacity span several orders of magnitude (Co-

hen, 1995; van den Bergh and Rietveld, 2004), some authors reject the notion for humans altogether (Simon, 1996), and others let the ceiling itself vary logistically (Meyer, 1994; Meyer and Ausubel, 1999) or treat natural capital as a depreciating asset (Dasgupta, 2021). Demographic projections used in most global assessments assume that population unfolds independently of environmental constraints (United Nations, 2022). Within this literature the environment has one book; the question of what an aggregate over two books with different yields and regeneration times can conceal has not been posed as a modeling problem, although the land-use literature has documented the historical operator – conversion of forest, grassland and wetland into cropland – in detail (Haberl et al., 2007; Krausmann et al., 2013; Erb et al., 2017).

Consider an orchard with two kinds of ground. Fast provisioning land – cropland and intensively managed systems – yields much and is quickly restored, but has little standing regenerative value. Ecological-capital land – forest, soil, wetland, long-rotation systems – has a large standing value and a regenerative flow that is slowly rebuilt. Demand is evaluated against the human-available annual account total; any remaining account shortfall triggers conversion from capital land into fast provision. Because aggregate biocapacity is a value-weighted sum over the two land books, conversion can raise the account total while capital-land area A_c shrinks. A harvest that looks stable, or a rising aggregate measure of capacity, can therefore mask a shrinking ecological-capital account. Neither the orchard nor its harvesters respond instantly – forest loss, soil degradation and demographic adjustment unfold over decades – which is why societies can prosper for long periods while quietly undermining their productive base, only to decline abruptly (Tainter, 1988; Diamond, 2005). The companion ledger work calls the broader phenomenon the *productivity illusion* (Abaee, 2026c); this paper isolates a distinct compositional mechanism: conversion can reweight the books so that the aggregate rises while ecological-capital area falls, even with the technology wave off.

We address three questions. **(Q1)** Under what conditions does conversion of ecological-capital land raise the aggregate while capital-land area A_c falls, how long does the mask last, and can the composition be identified from the aggregate? **(Q2)** Is the resulting capital drawdown a catastrophic shift in the sense of the alternative-stable-states literature (Scheffer et al., 2001), with a fold, hysteresis and early-warning signals, or something structurally different? **(Q3)** Is the mask self-limiting, and what does recovery require?

The paper makes four contributions. First, it formulates a minimal two-

land model (Section 2) in which area is conserved one-to-one across three land books, per-hectare capital quality is a separate state, conversion is a rate with an explicit time constant, and degradation acts on the surviving yield through an accumulated ecological debt; the model is a delay differential equation with ecological (regeneration) and demographic delays (Kuang, 1993; Hutchinson, 1948). Second, it proves three propositions (Section 3): a composition identity that gives the exact sign condition for the conversion contribution and shows that an aggregate-only observation does not locally identify the capital loss (Proposition 1); the uniqueness of the frozen-demand balance point, the location of the demand ceiling, and the absence of a fold and of critical slowing down, derived from the affine structure of the deficit-side dynamics (Proposition 2); and a boundedness asymmetry between the technology and debt channels that makes the mask transient (Proposition 3). Third, it reports NFA Edition 2026 world account movements for 1961–2023, with a same-edition 1961–2022 sensitivity, and separates these accounting results from physical-stock evidence (Section 3.5). A supplementary area–cereal-yield proxy is explicitly exploratory: its residual is an accounting construction, not ecological capital. Fourth, it states the falsifiable content of the model as five predictions and draws the reporting implication for sustainability indicators (Sections 3.6 and 4). The model is deliberately simple and most of its parameters are representative rather than fitted; it is not a forecast. Its purpose is to make the composition logic of aggregate biocapacity explicit and to state which outcomes are structural and which are conditional on effective parameters that are not separately identified.

2. Methods

2.1. A two-land model with conserved area and a separate quality state

The bioproductive area A_{tot} is divided into three books: fast-provisioning land A_f (cropland and intensively managed systems), ecological-capital land A_c (forest, soil, wetland and long-rotation systems, consolidated into one effective class), and a reserve A_r (fallow, retired or not yet allocated land), with $A_f + A_c + A_r = A_{tot}$ at all times. Capital land carries a per-hectare quality index $q \in [0, q_{\max}]$ (a capacity state, not an area), which regenerates logistically,

$$G_c(q) = \rho_c \, q \left(1 - \frac{q}{q_{\max}}\right), \tag{1}$$

so that the per-hectare value of capital land, in annual NFA-account-yield units, is

$$\Upsilon_c(q) = b_c + b_{G,c}\, G_c(q), \tag{2}$$

the sum of a direct per-hectare yield b_c and the regeneration-value coefficient $b_{G,c}$ times the quality growth $G_c(q)$. Biocapacity is the additive two-book aggregate

$$B = Y_f + Y_c = b_f A_f + \Upsilon_c(q)\, A_c, \tag{3}$$

in gha, the annual NFA account unit (Global Footprint Network, 2009, 2019). Demand is the footprint $E = eP$ of a population P with constant per-capita footprint e (gha cap^{-1}), and the shortfall above the human-available amount is

$$S = \left[E - \sigma_f Y_f - \sigma_c Y_c\right]_+ , \tag{4}$$

with $[x]_+ = \max(x, 0)$ exact (never smoothed) and $\sigma_f = \sigma_c = 1$ in the baseline. The shortfall is closed by converting capital land into fast land at a finite rate,

$$u_c = \frac{1}{\tau_{\text{conv}}} \min\!\left(\frac{S}{b_f},\; A_c - A_c^{\min}\right) \qquad [\text{ha yr}^{-1}], \tag{5}$$

where S/b_f is the area that would close the shortfall at the fast-land yield, $A_c^{\min}$ is a typed viability floor, and τ_{conv} is the conversion time constant. The land books evolve by one-to-one area transfers only,

$$\dot{A}_c = -u_c + R_{rc}, \tag{6}$$

$$\dot{A}_f = u_c + \mu A_r - \eta_f A_f, \tag{7}$$

$$\dot{A}_r = -\mu A_r + \eta_f A_f - R_{rc}, \tag{8}$$

where μA_r is cropland maintenance (replanting from the reserve), $\eta_f A_f$ is fast-land retirement to the reserve, and R_{rc} is an explicit, surplus-gated restoration flow from reserve to capital land,

$$R_{rc} = \chi_r\, A_r\, \left[B - E\right]_+ , \tag{9}$$

which is the only route back into the capital book. Equations (6)–(8) sum to zero, so the reserve is a constrained balance rather than an independent state. Quality regenerates with a recruitment delay, population follows a delayed logistic law (Pearl, 1925; Hutchinson, 1948) on the carrying capacity

$K = B/e$, and degradation acts on the surviving fast-land yield through an accumulated ecological debt D:

$$\dot{q} = G_c\big(q(t-\tau_g)\big), \tag{10}$$

$$\dot{P} = rP\left(1 - \frac{P}{K(t-\tau_p)}\right), \qquad K = \max(B/e,\, K_{\min}), \tag{11}$$

$$b_f = \big(b_{f0} + T_b(t)\big)\, e^{-\alpha D}, \tag{12}$$

$$\dot{D} = [E - B]_+ - \eta D, \tag{13}$$

$$T_b(t) = \omega_{Tb}\, \frac{\Delta b}{1 + e^{-\kappa_w (t - t_{wave})}}, \qquad \omega_{Tb} \in \{0, 1\}. \tag{14}$$

Because model time t is measured in years and $E - B$ is an annual account amount in gha, the leaky shortfall accumulator D has units gha yr: both $\dot{D}$ and ηD have units gha. This is a model-internal time-integrated liability, not an NFA-reported footprint or biocapacity item.

The technology wave T_b is a bounded logistic yield step controlled by ω_{Tb}: $\omega_{Tb} = 0$ in the baseline, main figures, and all tables except the separately identified matched experiment in SI §S8, where $\omega_{Tb} = 1$. The state is (A_f, A_c, q, P, D) with one delay each in regeneration and demography; a timber-liquidation channel and a direct fast-to-capital restoration flow are stated extensions that remain off in every reported run (Supplementary Information, SI §S1).

Five conventions are essential. (i) *The positive part is exact.* A smoothed ramp would leak conversion when $E < B$; the one-sided deficit mechanism is the model's content, so $[\cdot]_+$ is never regularized and stability at $S = 0$ is one-sided. (ii) *Conversion is a rate.* S/b_f is a stock (ha); it becomes a rate only through τ_{conv}, so u_c is dimensionally $\text{ha}\,\text{yr}^{-1}$ and conversion is never instantaneous. (iii) *Area is conserved one-to-one and capacity is a separate book.* Every area flow in (6)–(8) moves hectares between books; regeneration G_c is a flow of capacity into q and never transfers land, so conservation is a property of the equations, not an identity imposed afterwards. (iv) *Annual account amounts and their time-integral are never interchanged.* B, E, Y_f, Y_c and S are annual NFA account amounts in gha; D is a leaky time-integral of shortfall in gha yr, repaid at rate η; b_f, b_c are in $\text{gha}\,\text{ha}^{-1}$ and $b_{G,c}$ is a regeneration-value coefficient in $\text{gha}\,\text{yr}\,\text{ha}^{-1}$. (v) *A sustainable cropland equilibrium exists.* Because (7) contains the maintenance term μA_r, a positive fast book A_f^* is sustained without conversion; without it the fast book would decay to zero at every equilibrium.

Two positive feedback loops act on different books. The *conversion loop* reallocates area: $A_c \downarrow\rightarrow A_f \uparrow\rightarrow$ a positive conversion contribution to B (if $b_f > \Upsilon_c$) $\rightarrow K \uparrow\rightarrow P \uparrow\rightarrow S \uparrow\rightarrow A_c \downarrow$; other terms can alter the net change in B, but this loop is the composition mechanism of the paper. The *debt loop* erodes the surviving yield: $D \uparrow\rightarrow b_f \downarrow\rightarrow B \downarrow\rightarrow S \uparrow\rightarrow D \uparrow$; it later turns against the aggregate and is one reason the mask is bounded (Proposition 3).

2.2. Parameters and their status

Table 1 lists the baseline registry. The capital book is a strategic reduction: ecological capital is physically heterogeneous, with land types that differ in extent, condition, stock, yield, and regeneration time, while the model represents them as one effective class. Most registry values are therefore representative, effective quantities, not fitted estimates. Independent recovery data bound a broad regeneration-response timescale band across secondary forests, collapsed fish stocks, and soils (Poorter et al., Poorter et al., 2016; RAM Legacy, cf. Hutchings and Reynolds, 2004; Neubauer et al., 2013; Poeplau et al., 2011; SI §S5.5); these anchors bound a broad effective timescale band of about 25–33 yr (representative value about 29 yr; forest $t_{50} \approx 30$ yr, fishery $t_{50} \approx 5.6$ yr). They concern regeneration response, not the delay τ_g, and do not identify a single global rate. The NFA cropland biocapacity-per-area quotient and a supplementary FAOSTAT cereal-yield/cropland-area proxy are descriptive account diagnostics, not fitted values of model b_f or a physical-capital observation operator (SI §S5). The external proxy uses an explicit 1961 NFA account anchor, but its cereal-yield and area boundaries do not constitute a validated mapping to the model's fast book. The baseline sits in the composition-premium regime, $\Delta b_{\text{conv}} = b_f - \Upsilon_c(q) > 0$, which permits a positive conversion contribution (Section 3.1); the registry fixes a regime, not a prediction.

2.3. Data and the observation operator

The empirical analysis uses the official National Ecological Footprint and Biocapacity Accounts, Edition 2026, results and metadata version 1.0 (Lo et al., 2026). The primary window is the complete world series 1961–2023; rows for 2024–2025 are estimated and excluded. The 1961–2022 sensitivity is taken from the same 2026 edition, not from a different vintage. Edition 2026 recalculates common years and revises several source inputs and methods, so earlier-edition totals are not blended with the current series. From the workbook's `world_data` sheet, B is `biocap_total_gha`; the empirical

Table 1: Baseline parameter registry (single source of truth for all reported runs) and the status of each parameter. "Representative" values are effective, lumped quantities that are not individually identified; "anchored" values are bounded by independent data (SI §S5); "extension" parameters are stated but off in the base model. Full symbol table with units in SI §S1.

Parameter	Value	Meaning	Status
ρ_c	$0.08\,\mathrm{yr}^{-1}$	quality regeneration rate of capital land	representative; timescale band anchored
b_{f0}	$0.85\,\mathrm{gha\,ha}^{-1}$	base fast-land yield per hectare	representative (2026 NFA account value/area quotient ≈ 1.045 in 1961; not a physical-yield measure)
b_c	$0.05\,\mathrm{gha\,ha}^{-1}$	capital-land direct yield	representative
$b_{G,c}$	$0.8\,\mathrm{gha\,yr\,ha}^{-1}$	regeneration-value coefficient for capital land	representative
$q_{\max}$	1.0 (dimensionless)	normalized quality ceiling	normalization
μ	$0.06\,\mathrm{yr}^{-1}$	cropland maintenance (reserve→fast)	representative
η_f	$0.05\,\mathrm{yr}^{-1}$	fast-land retirement (fast→reserve)	representative
τ_{conv}	1.0 yr	conversion time constant	representative
α	$0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	degradation of b_f per unit debt	representative
η	$0.05\,\mathrm{yr}^{-1}$	debt-repayment rate (> 0)	representative
e	$0.55\,\mathrm{gha\,cap}^{-1}$	per-capita footprint (constant)	representative
r	$0.02\,\mathrm{yr}^{-1}$	intrinsic population growth	representative
$\tau_g,\ \tau_p$	20, 25 yr	regeneration and demographic delays	representative; τ_p lumps institutional lags
A_{tot}	2.5 ha	total bioproductive area	normalization
$\sigma_f,\ \sigma_c$	1, 1 (dimensionless)	human-available shares of annual account amounts	baseline (< 1: reservation extension)
χ_r	$0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	surplus-gated restoration coefficient	on with endogenous demand; off under frozen demand
$A_c^{\min}$, $A_r^{\min}$, $K_{\min}$	0.05 ha, 0.10 ha, 0.02 cap	typed floors	domain constraints
Δb, κ_w, t_{wave}	$1.0\,\mathrm{gha\,ha}^{-1}$, $0.1\,\mathrm{yr}^{-1}$, 40 yr	technology wave	off in baseline; active only in matched SI S8 experiment

footprint series is `efp_total_gha`, the Global Ecological Footprint of Production, denoted E_{prod}; and population is `pop_1000_persons`, defined in the workbook as UN PopStat population. Per-capita quantities are recomputed by dividing each total account value by that population in persons, rather than using the rounded workbook per-capita columns. Ratios, indices, and logarithmic changes are dimensionless.

The component series retain the workbook labels Cropland, Grazing land, Forest biocapacity, Fishing grounds, and Built-up land. These are NFA account values, not measurements of biomass, soil condition, wetland extent, fish biomass, or the model's effective area A_c. Built-up land is an NFA accounting class and is not ecological capital. Accordingly, the land-type movements are used as accounting-composition signals only. The workbook, extracts, field definitions, and reproducible scripts are described in the package data notes and SI §S5.

A supplementary proxy sensitivity uses OWID's FAOSTAT-derived series for world cropland area (arable land plus permanent crops) and cereal yield as separate physical inputs (Our World in Data, 2026a,b). The cereal series is a sentinel, not an all-crop yield measure or an NFA yield factor. Its base-year account level is anchored explicitly to 1961 NFA cropland biocapacity; the proxy $X_t = \hat{b}_{f,t} A_{f,t}$ and residual $C_t = B_t - X_t$ are therefore an assumed observation operator, not a measured partition of physical capital. C is an accounting residual only and cannot be identified with $\Upsilon(q)A_c$ without independent, boundary-consistent inputs and a declared mapping and valuation. The proxy and its calibration sensitivity are reported in SI §S5; no physical-stock trend is inferred from it.

The aggregate-versus-per-capita comparison uses the exact identity $\Delta \ln B = \Delta \ln(B/P) + \Delta \ln P$. This extensive/intensive split is an accounting identity, not a causal decomposition of biocapacity production.

2.4. Numerical solution and verification

The delay system is integrated by the method of steps with an explicit first-order Euler scheme ($\Delta t = 0.05\,\mathrm{yr}$ for the demonstration run, 0.1 and $0.05\,\mathrm{yr}$ for the drawdown sweeps). The exact positive part is used; state bounds enforce $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P, D \geq 0$, and $K \geq K_{\min}$. The constant pre-run histories of B and q cover the maximum active delay; the initial B sample is computed from the initial state, not set to zero. In the SI wave experiment, $T_b(t)$ is applied consistently to the current fast-land yield, returned $B(t)$, and the delayed B history used in $K(t - \tau_p)$; the

base-model wave remains off. The corrected history initialization and consistent wave bookkeeping avoid an artificial time-zero capacity dip in the delayed carrying-capacity series. Area conservation is checked at every step ($\max|A_f + A_c + A_r - A_{tot}| = 2.2 \times 10^{-16}$ ha). Every analytic quantity below (balance points, ceiling, eigenvalues, composition diagnostic) is recomputed and asserted before the figure is written. Drawdown thresholds are reported as bands because the operational floor-proximity criterion depends on tolerance and step size (Section 3.2). The solver used for the main trajectory evaluates the regeneration term of (10) at current rather than delayed quality; because q is monotone and the regenerative term is small at baseline ($b_{G,c}G_c \leq 0.016\,\mathrm{gha\,ha^{-1}}$ versus $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$), applying the delay changes the reported quantities by less than 1% (window 149 versus 150 yr; initial-population thresholds within 0.001 cap), and both variants are documented in SI §S3. The controlled wave run and step-halving checks are in SI §S8. The v44 predecessor package remains public; the current v45 working package, including the model code, NFA Edition 2026 reanalysis, source inputs, and generated figures, is described in the Data availability statement and awaits the version, citation, and build checks before public deposit.

3. Results

3.1. The composition mask and what the aggregate cannot identify

The per-hectare value of capital land $\Upsilon_c(q)$ is a concave quadratic in q with an interior maximum at $q^* = q_{\max}/2$, where $\Upsilon_c(q^*) = b_c + b_{G,c}\rho_c q_{\max}/4 = 0.066\,\mathrm{gha\,ha^{-1}}$ at the baseline; the regenerative term there ($0.016\,\mathrm{gha\,ha^{-1}}$) is about one third of the direct yield ($0.05\,\mathrm{gha\,ha^{-1}}$; ratio 0.32), so capital value is only modestly enhanced by regeneration. The quantity that governs what conversion does to the aggregate is the *composition diagnostic*

$$\Delta b_{\mathrm{conv}}(q, D) = b_f(D) - \Upsilon_c(q), \tag{15}$$

the difference between what a hectare yields in the fast book and what it is worth in the capital book. At the baseline $\Delta b_{\mathrm{conv}} = +0.787,\ +0.784,\ +0.794\,\mathrm{gha\,ha^{-1}}$ at $q = 0.3, 0.6, 0.9$: the conversion contribution is robustly positive, although other terms still determine the net account change. The three regimes $\Delta b_{\mathrm{conv}} > 0$ (premium: conversion contributes positively to B, so a conversion-driven mask is possible), $\Delta b_{\mathrm{conv}} = 0$ (first-order neutrality, non-generic) and $\Delta b_{\mathrm{conv}} < 0$ (discount: conversion

contributes negatively to B, so conversion alone cannot produce that mask) are the only shape distinction the quality-based model admits; because Y_c is linear in A_c at fixed q, there is no area-based capacity maximum and no regime split of the kind an area-based regeneration law would produce.

Proposition 1 (Composition identity, mask and non-identifiability)**.** *Let $T_b \equiv 0$, $R_{rc} \equiv 0$ and hold D fixed, so that b_f is constant. Along any solution of (3)–(10),*

$$\dot{B} = \Delta b_{\text{conv}}(q)\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}\, G_c'(q)\, A_c\, \dot{q}. \tag{16}$$

(i) Whenever $u_c > 0$, the conversion contribution to $\dot{B}$ is $\Delta b_{\text{conv}}(q)u_c$ and is positive if and only if $\Delta b_{\text{conv}}(q) > 0$; with $R_{rc} = 0$, the same transfer gives $\dot{A}_c = -u_c < 0$. At any state where the maintenance–retirement imbalance and quality-growth terms vanish ($\mu A_r = \eta_f A_f$ and $\dot{q} = 0$), the net derivative is $\dot{B} = \Delta b_{\text{conv}}(q)u_c$, so its sign has this same condition.

(ii) In the reduced frozen-demand, fixed-D case in which $\mu A_r = \eta_f A_f$ and $\dot{q} = 0$ throughout, and with $\sigma_f = \sigma_c = 1$, $\Delta b_{\text{conv}} > 0$ implies $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ whenever $S > 0$. Thus the shortfall decreases to zero if the balance is reached first; if the floor constraint becomes active first, A_c approaches $A_c^{\min}$ asymptotically and the shortfall can persist. This monotonicity is not asserted when the other terms in the full model are nonzero.

(iii) At fixed $b_f > 0$ and fixed q (so that $\Upsilon_c(q) > 0$), the observation $B = b_f A_f + \Upsilon_c(q) A_c$ is one scalar constraint on two land areas. At an interior feasible state it leaves a local continuum of decompositions: a sufficiently small δA_c can be offset by $\delta A_f = -(\Upsilon_c/b_f)\delta A_c$, with $\delta A_r = -\delta A_f - \delta A_c$, while preserving B and the area total, provided the area bounds remain satisfied. Thus an aggregate-only observation does not locally identify A_c. For a finite observed change ΔB, the algebraic offset $\Delta A_f = (\Delta B - \Upsilon_c \Delta A_c)/b_f$ is feasible only when the resulting land areas respect their bounds; the admissible range depends on the state and need not be the whole interval from $A_c^{\min} - A_c(0)$ to zero. If b_f and A_f are independently observed, the residual $C = B - b_f A_f = \Upsilon_c(q) A_c$ identifies that product, not Υ_c and A_c separately. An independent observation of A_c would identify the effective value $\Upsilon_c(q) = C/A_c$; even then, b_c, $b_{G,c}$,

ρ_c, and q are not separately identified from a single value. With q independently observed over a sufficiently informative range, $\Upsilon_c(q)$ can at most identify b_c and the product $b_{G,c}\rho_c$, not $b_{G,c}$ and ρ_c separately.

Proof. Differentiate (3) with b_f constant and substitute (6)–(7) with $R_{rc} = 0$: $\dot{B} = b_f(u_c + \mu A_r - \eta_f A_f) - \Upsilon_c u_c + b_{G,c} G_c'(q) A_c \dot{q}$, which is (16). (i) Follows from (16) with $R_{rc} = 0$; when the maintenance–retirement imbalance and quality-growth terms vanish, only the conversion term remains. (ii) Under the reduced-case conditions in the statement, $S = E - B$ and $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ on $S > 0$. Since S is bounded below by zero and A_c is bounded below by $A_c^{\min}$, conversion either closes the shortfall or is limited by the floor. (iii) At fixed b_f and q, the observation map is linear; an interior feasible point admits the stated small perturbations along a level set of B, while finite changes must also obey the area bounds. The residual C identifies only $\Upsilon_c(q)A_c$ unless A_c is independently observed; the parameter combination follows from the definition of $\Upsilon_c(q)$. □

Figure 1a shows the mask at the baseline under frozen demand $E = 0.825\,\text{gha}$ (restoration off, $A_{f0} = 0.6\,\text{ha}$, $A_{c0} = 1.5\,\text{ha}$, dimensionless $q_0 = 0.9$, $A_r(0) = 0.4\,\text{ha}$). Biocapacity rises from 0.594 to 0.825 gha (+39%) while capital land area A_c falls from 1.500 to 0.808 ha (−46%) and fast land A_f rises from 0.600 to 0.923 ha. The dimensionless quality state q rises from 0.9 to 1.0, but because $q_0 = 0.9 > q^* = q_{\max}/2 = 0.5$, this rise lies on the descending side of the concave per-hectare function $\Upsilon_c(q)$: its value falls from 0.05576 to 0.05000 gha ha^{-1}. Accordingly, the modeled capital contribution $\Upsilon_c(q)A_c$ to B falls from 0.08364 to 0.04038 gha (by 51.7%, about 52%), distinct from the 46% decline in capital-land area. The area changes are one-to-one ($\Delta A_f + \Delta A_c + \Delta A_r = 0$ to machine precision). The dynamics are fast at first and then asymptotic: because the deficit-side land subsystem is affine (Proposition 2(ii)), the approach to the balance point is exponential at the slow rate $|\lambda_1| = 0.054\,\text{yr}^{-1}$ (e-folding $\approx 19\,\text{yr}$), so within 25 yr 98% of the rise in B and 85% of the fall in A_c have occurred ($B = 0.820\,\text{gha}$, $A_c = 0.913\,\text{ha}$; 97% of the fall by 50 yr), while the strict inequalities $\dot{B} > 0$ and $\dot{A}_c < 0$ persist for $\approx 150\,\text{yr}$ at numerical resolution. The slow tail is reserve equilibration: retirement of fast land to the reserve lowers the fast flow, re-opens a small shortfall, and converts a little more capital, until the reserve reaches its equilibrium share $A_r/A_f = \eta_f/\mu$. The window is therefore not a fixed horizon: its scale is set by λ_1, which depends on τ_{conv}, μ, η_f and

the yield ratio b_c/b_f (SI §S2.2), and its amplitude by Δb_{conv} and the initial deficit.

Two consequences of Proposition 1(iii) frame the empirical analysis of Section 3.5. First, the composition signature (B rises while A_c falls) is contemporaneously observable only in the two-book series; a monitor who sees the aggregate alone cannot distinguish a composition mask from broad growth. Second, independent proxies for the fast book (A_f, b_f) convert the aggregate into a measured fast book and a residual, which is the most that can be identified without independent ecological valuation of the capital book. We call the net two-book signature B rising while A_c falls the *composition mask*; its conversion component is positive when $\Delta b_{\text{conv}} > 0$, but the net sign of $\dot{B}$ also depends on the other terms in (16). The alternative *liquidation mask* is E steady while A_c falls, which does not require a positive conversion contribution. The two-book series makes these readings distinguishable.

3.2. The drawdown is a demand-driven floor collision, not a fold

With endogenous population there is no interior steady state with active conversion: at a demographic equilibrium $P = K = B/e$, so $E = B$ and $S = 0$. Conversion is therefore a transient regime, and its steady states exist only under a frozen exogenous demand pinned above the flow. The frozen-demand system is where the question of a catastrophic shift must be posed, and it has a complete answer.

Proposition 2 (Balance point, demand ceiling and affine deficit-side dynamics)**.** *Assume frozen demand E, $T_b \equiv 0$, $R_{rc} \equiv 0$, $q = q_{\max}$ (the attractor of (10) for $q(0) > 0$) and $D = 0$. Let $k = \mu/(\mu + \eta_f)$ and*

$$B_{eq}(A_c) = b_{f0}\,k\,(A_{tot} - A_c) + b_c A_c = b_{f0} k A_{tot} - (b_{f0}k - b_c)A_c. \tag{17}$$

(i) Equilibria of the land subsystem on the no-conversion face $S = 0$ satisfy $A_f = k(A_{tot} - A_c)$, $A_r = (1-k)(A_{tot} - A_c)$ and $B_{eq}(A_c) \geq E$. If $b_{f0}k > b_c$, B_{eq} is strictly decreasing, so the balance point $B_{eq}(A_c^*) = E$ is unique,

$$A_c^*(E) = \frac{b_{f0}kA_{tot} - E}{b_{f0}k - b_c}, \tag{18}$$

The algebraic root lies in the feasible land interval $A_c \in [A_c^{\min}, A_{tot}]$ exactly when $B_{eq}(A_{tot}) \leq E \leq E_{\text{ceil}} := B_{eq}(A_c^{\min})$; it lies strictly above $A_c^{\min}$ iff $E < E_{\text{ceil}}$. For $E < B_{eq}(A_{tot})$, the root lies above A_{tot} and feasible no-conversion equilibria already satisfy $B_{eq} \geq E$.

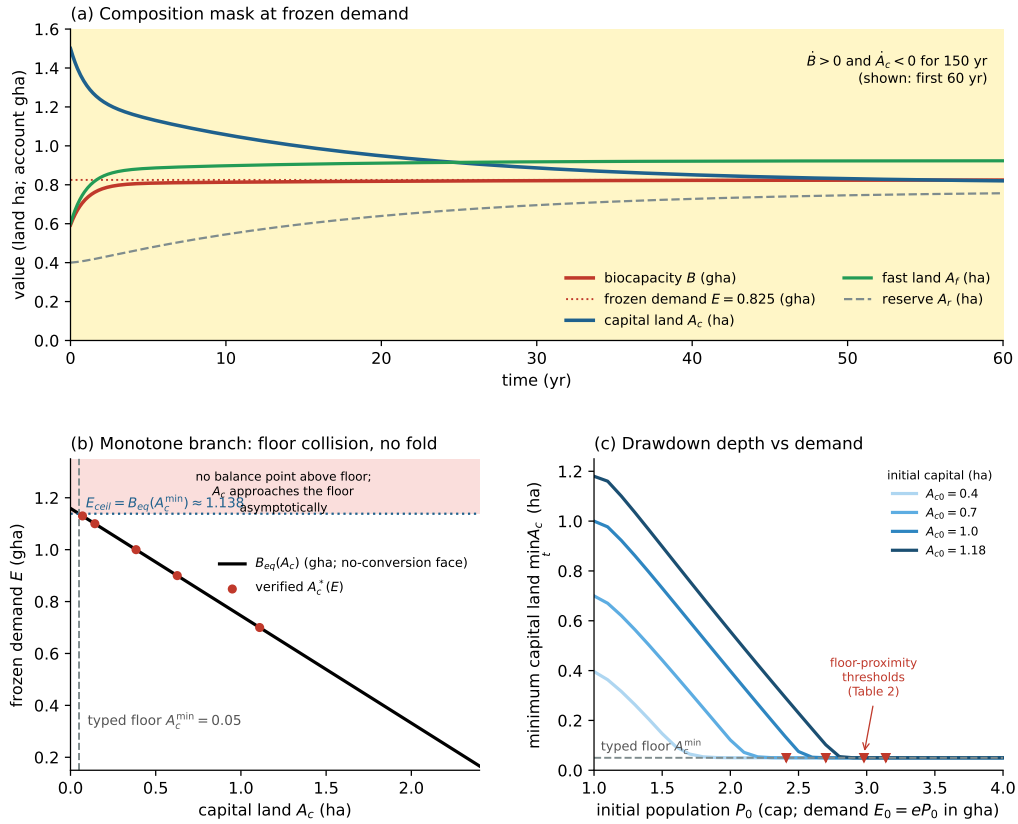


Figure 1: Model dynamics at the baseline registry (Table 1), regenerated from the deposited code. Account amounts B and E are in gha; land areas A_f, A_c, A_r are in ha; population P_0 is in cap. (a) Composition mask under frozen demand $E = 0.825$ (restoration off, $A_{f0} = 0.6$, $A_{c0} = 1.5$, dimensionless $q_0 = 0.9$): biocapacity B rises from 0.594 to 0.825 while capital land A_c falls from 1.500 to 0.808 and fast land A_f rises from 0.600 to 0.923; the shaded region marks the interval on which $\dot{B} > 0$ and $\dot{A}_c < 0$ hold simultaneously (≈ 150 yr at numerical resolution; only the first 60 yr are shown; 98% of the rise in B and 85% of the fall in A_c occur within the first 25 yr). (b) The frozen-demand balance branch $B_{eq}(A_c)$ of Proposition 2, strictly decreasing in A_c, with the verified balance points $A_c^*(E) = 1.110, 0.626, 0.385, 0.143, 0.070$ at $E = 0.70, 0.90, 1.00, 1.10, 1.13$; the branch meets the typed floor $A_c^{\min} = 0.05$ at the demand ceiling $E_{\text{ceil}} \approx 1.138$, above which no admissible balance point exists and the capital book approaches the floor (shaded). (c) Endogenous-demand runs ($\tau_g = 20$, $\tau_p = 25$ yr, $\chi_r = 0.10$): the minimum capital land reached during a run is a continuous, monotone function of initial demand P_0 (hence $E_0 = eP_0$) for each initial capital A_{c0}; the floor is approached asymptotically, and the floor-proximity thresholds of Table 2 (triangles) are the operational points at which the floor is approached to within 10^{-6} ha. There is no discontinuity in the outcome as a function of demand.

(ii) On the deficit side ($S > 0$, $S/b_f < A_c - A_c^{\min}$) the land subsystem is affine in $z = (A_f, A_c)$, $\dot{z} = Jz + c$, with

$$J = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix}, \qquad \det J = \frac{\mu + \eta_f}{\tau_{\text{conv}}} \Big(k - \frac{b_c}{b_f} \Big), \tag{19}$$

and $\operatorname{tr} J < 0$. J does not depend on E or on the state, $\det J > 0$ exactly when B_{eq} is decreasing, and the fixed point of the affine system is the balance point of (i). For $B_{eq}(A_{tot}) \le E < E_{\text{ceil}}$ the balance point lies above the typed floor and every trajectory that remains in this regime converges to it at the constant rates given by the eigenvalues of J; for $E > E_{\text{ceil}}$ the affine fixed point lies below the floor, so the trajectory approaches the face $A_c = A_c^{\min}$ asymptotically, with a limiting persistent shortfall $S^* = E - E_{\text{ceil}} > 0$. The ceiling is a collision of this reduced equilibrium relation with a face of the admissible set, not a bifurcation of its vector field: there is no second branch or fold, no hysteresis of this frozen-demand equilibrium relation, and the leading eigenvalue does not tend to zero as $E \to E_{\text{ceil}}$. These statements apply under the assumptions above; they are not claims that the realized land allocation is independent of its history.

(iii) Beyond the ceiling, with degradation active, the debt converges to the unique root $D^*(E)$ of $\eta D = E - B_{eq}(A_c^{\min}; D)$, where $B_{eq}(\cdot\,; D)$ uses $b_{f0} e^{-\alpha D}$, provided $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min})$; D^* is increasing in E.

Proof. (i) With $S = 0$ and $R_{rc} = 0$, (7) gives $\mu A_r = \eta_f A_f$, hence $A_r/A_f = \eta_f/\mu$ and, with $A_r = A_{tot} - A_f - A_c$, $A_f = k(A_{tot} - A_c)$; substituting into (3) with $q = q_{\max}$ ($\Upsilon_c = b_c$) and $D = 0$ gives $B = B_{eq}(A_c)$, and $S = 0$ requires $B \ge E$. Strict monotonicity and (18) follow from $b_{f0} k > b_c$. (ii) On the deficit side $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$ is affine in z; substituting into (6)–(7) with $A_r = A_{tot} - A_f - A_c$ gives J and a constant vector c by direct differentiation; $\det J$ follows by expansion, and $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f < 0$ for $b_c < b_f$. The fixed point solves $u_c = 0$ and $\mu A_r = \eta_f A_f$, i.e. it is the balance point. Because J is constant, convergence rates are constant along the branch. (iii) The map $D \mapsto \eta D + B_{eq}(A_c^{\min}; D) - E$ is continuous, negative at $D = 0$ when $E > E_{\text{ceil}}$, tends to $+\infty$, and is strictly increasing

when $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min}) \geq -\partial_D B_{eq}$; hence it has a unique root, which increases with E. □

The uniqueness result is about the reduced equilibrium relation, not path independence of realized states. In the baseline trajectory, $R_{rc} = 0$ and the conversion flow u_c has no reverse counterpart. When the shortfall closes, A_c stops declining and approaches its lower balance value; it remains below its pre-overshoot level because this run contains no restoration flow. The quality state q evolves separately and can continue changing $\Upsilon_c(q)$ even as A_c settles. Thus the model has a unique frozen-demand balance point under Proposition 2's restrictions, but a history-dependent post-overshoot land allocation; this is not perpetual decline at equilibrium and is not a general claim about real ecosystems.

At the baseline $k = 0.5455$, the slope $b_{f0}k - b_c = 0.414\,\mathrm{gha\,ha^{-1}}$, and $E_{\mathrm{ceil}} = 1.138\,\mathrm{gha}$; the balance points $A_c^*(E) = 1.110, 0.626, 0.385, 0.143, 0.070\,\mathrm{ha}$ at $E = 0.70, 0.90, 1.00, 1.10, 1.13\,\mathrm{gha}$ agree with the integrated equilibria (Fig. 1b). The eigenvalues of J are real and negative, $\lambda_1 = -0.0537\,\mathrm{yr^{-1}}$ and $\lambda_2 = -0.998\,\mathrm{yr^{-1}}$; λ_1 varies only weakly with the fast-land yield, from $-0.0522\,\mathrm{yr^{-1}}$ at $b_f = 0.60$ to $-0.0556\,\mathrm{yr^{-1}}$ at $b_f = 2.00$ (Table 2a), and never changes sign. There is no universal scalar stability index: the conversion-loop gain is the state- and parameter-dependent Δb_{conv}, of order $1\,\mathrm{gha\,ha^{-1}}$, and the return rate is $|\lambda_1|$ of order $10^{-2}\,\mathrm{yr^{-1}}$. Beyond the ceiling the debt fixed points are $D^* = 1.19$ and $5.38\,\mathrm{gha\,yr}$ at $E = 1.158$ and $1.238\,\mathrm{gha}$ (integrated values 1.21 and $5.39\,\mathrm{gha\,yr}$); the monotonicity condition of (iii) holds with margin ($\alpha b_{f0} k (A_{tot} - A_c^{\min}) = 0.034\,\mathrm{yr^{-1}} < \eta = 0.05\,\mathrm{yr^{-1}}$), so under frozen demand the debt saturates rather than running away.

The statement "no critical slowing down" is precise here. By critical slowing down we mean that the least-negative eigenvalue of the relevant linearization tends to 0^- as a parameter approaches a fold, so that the return time $T_{ret} \approx -1/\lambda_1$ diverges. In Proposition 2 the linearization is constant along the entire branch, so neither λ_1 nor $T_{ret} \approx 19\,\mathrm{yr}$ changes as $E \to E_{\mathrm{ceil}}$. The only early-warning signal the model offers is a *boundary* signal, the capital book approaching its typed floor, which is observable only in the two-book series.

With endogenous demand the same structure appears as a continuous drawdown. Figure 1c shows the minimum capital land reached during a run as a function of initial population P_0 for four initial capital-land ar-

eas A_{c0}: the curves are continuous and monotone, the floor is approached asymptotically (because $u_c \leq (A_c - A_c^{\min})/\tau_{\text{conv}}$, the distance to the floor decays exponentially during a sustained deficit and is never crossed), and higher initial capital-land area supports higher demand before the floor is approached. The "collapse thresholds" of the earlier literature on this model are therefore operational quantities: Table 2b reports the initial demand at which the floor is approached to within 10^{-2}, 10^{-4} and 10^{-6} ha of its value. The 10^{-6} ha floor-proximity thresholds ($P_0 \approx 2.41, 2.70, 2.98, 3.14$ cap for $A_{c0} = 0.40, 0.70, 1.00, 1.18$ ha) are obtained by bisection on that tolerance criterion; they rise monotonically with initial capital, as do the looser criteria, and no criterion exhibits a jump. Two robustness checks bear on the predictions of Section 3.6: the initial-population floor-proximity threshold at $A_{c0} = 1.0$ ha is identical to three decimals for $\tau_g = 5$, 20 and 40 yr, and identical to three decimals with the restoration coefficient $\chi_r = 0.10\,\text{gha}^{-1}\,\text{yr}^{-1}$ and $\chi_r = 0\,\text{gha}^{-1}\,\text{yr}^{-1}$ (SI §S4). The first reflects the small leverage of the regenerative term at the baseline and the monotone quality dynamics; the second reflects the fact that a surplus-gated flow is inert during the shortfall that produces the drawdown.

3.3. The mask is bounded: technology gain is parameter-bounded, debt erosion is shortfall-bounded

Both the technology wave and the degradation act on the same coefficient, the fast-land yield b_f of (12), and they do so multiplicatively. This yields a structural asymmetry.

Proposition 3 (Boundedness asymmetry)**.** *Under* (12)–(14)*:*

(i) the technology factor satisfies $1 \leq (b_{f0}+T_b(t))/b_{f0} \leq 1+\Delta b/b_{f0}$ *for all* t*, so the logarithmic gain of the yield channel is bounded by* $\ln(1+\Delta b/b_{f0})$ *uniformly in time (*0.78 *at the baseline);*

(ii) the debt satisfies $D(t) \leq \max\{D(0),\ \sup_{s\leq t}[E-B]_+(s)/\eta\}$*, and* $D \to S^*/\eta$ *whenever the shortfall settles at* S^**; the degradation factor* $e^{-\alpha D}$ *is therefore bounded below only through the sustained shortfall;*

(iii) the degradation loss exceeds the maximal technology gain once the sustained shortfall exceeds $S^\dagger = (\eta/\alpha)\ln(1+\Delta b/b_{f0})$*, equal to* 1.30 gha *at the baseline.*

Table 2: Model results at the baseline registry. (a) Composition diagnostic at $q^* = q_{\max}/2$ ($\Upsilon_c(q^*) = 0.066\,\mathrm{gha\,ha^{-1}}$), eigenvalues of the deficit-side matrix J of (19) and the demand ceiling, for a sweep of the fast-land yield b_f (all other parameters at baseline). (b) Initial population P_0 (cap; initial footprint $E_0 = 0.55P_0$ in gha) at which the minimum capital land reached during an endogenous-demand run ($A_{f0} = 0.6\,\mathrm{ha}$, dimensionless $q_0 = 0.9$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$) comes within the stated distance of the typed floor $A_c^{\min} = 0.05$; the first two columns are read from a P_0 grid of step 0.1 ($\Delta t = 0.1\,\mathrm{yr}$, $T = 400\,\mathrm{yr}$), the last is a bisection using the $10^{-6}\,\mathrm{ha}$ floor-proximity criterion ($\Delta t = 0.05\,\mathrm{yr}$, $T = 1000\,\mathrm{yr}$). Drawdown depth is continuous in P_0 (Fig. 1c); the thresholds are a band, not a single value.

(a) Fast-land yield sweep (all rows composition-premium, $\Delta b_{\mathrm{conv}} > 0$)

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.804
0.85 (baseline)	+0.784	−0.0537	−0.998	1.138
1.20	+1.134	−0.0547	−1.014	1.606
2.00	+1.934	−0.0556	−1.029	2.675

(b) Initial demand P_0 at which $\min_t A_c$ approaches $A_c^{\min}$ to within:

A_{c0} (ha)	within 10^{-2} ha	within 10^{-4} ha	within 10^{-6} ha
0.40	≈ 1.8	≈ 2.1	≈ 2.41
0.70	≈ 2.2	≈ 2.4	≈ 2.70
1.00	≈ 2.6	≈ 2.8	≈ 2.98
1.18	≈ 2.8	≈ 3.0	≈ 3.14

Under an additive degradation law $b_f = b_{f0} + T_b - \alpha D$ the two channels enter symmetrically and (iii) reduces to a comparison of Δb with αD^*; the asymmetry is a property of the multiplicative form.

Proof. (i) is $0 \le T_b \le \Delta b$. (ii) From (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S} = \sup_{s \le t} [E - B]_+$, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ by comparison; if $[E - B]_+ \to S^*$ then $D \to S^*/\eta$. (iii) compares $\alpha S^*/\eta$ with $\ln(1 + \Delta b/b_{f0})$. □

The proposition corrects a tempting overstatement: debt does not compound without bound under a fixed shortfall – it saturates at S^*/η – but its ceiling is set by the shortfall rather than by a parameter, and the shortfall is itself raised by demand growth. At the baseline the crossover shortfall $S^\dagger \approx 1.3$ gha exceeds the baseline biocapacity, so at these parameters the debt channel is a slow erosion rather than a reckoning; what bounds the mask at the baseline is the balance-point mechanism of Proposition 1(ii), with debt erosion as the second, slower limit. A second coupling is possible: because $K = B/e$, a technology-driven rise in B can raise carrying capacity and, after the demographic delay, demand. Whether this produces a larger all-time debt peak depends on the initial state and the timing of the response; it is not a sign-general theorem. In the controlled matched experiment in SI §S8, with $P_0 = 1.0$ cap, the active T_b run reaches a global D maximum of 0.02020 gha yr versus 0.00813 gha yr without the wave (ratio 2.49). At the reference $P_0 = 1.2$ cap, the all-time peak is instead lower with the wave (0.05104 versus 0.06122 gha yr), while the maximum over $t \ge t_{wave} = 40$ yr is higher (0.02518 versus 0.01085 gha yr; ratio 2.32). This delayed-rebound response is conditional on the declared parameters and initial demand, and the global maximum must not be conflated with the post-wave-center maximum. The active-wave experiment is a diagnostic extension, not a forecast or an empirical estimate.

3.4. Recovery requires an explicit, gated restoration flow

The model is structurally asymmetric: $A_c \to A_f$ proceeds through conversion, but no term in (6)–(8) returns land to the capital book except the restoration flow R_{rc}. Fast-land retirement sends land to the reserve, not to capital. Reducing population, footprint or shortfall therefore does not reverse prior conversion: with $\chi_r = 0$ the capital book stays where the shortfall left it. In the endogenous-demand run with $A_{c0} = 1.0$ ha and $P_0 = 2.0$ cap, capital land falls to 0.398 ha within four years and

remains there without restoration, whereas with the surplus-gated flow ($\chi_r = 0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$) it recovers to 0.828 ha, a dimensionless recovery fraction $\mathcal{R}_c = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}}) = 0.71$ (SI §S4.3). The sign of the gate matters. A surplus-gated flow, active when $B > E$, is the ecologically sign-correct rewilding: it retires land only when the system has spare capacity, but it is inert during the shortfall it must eventually address – which is why the drawdown thresholds of Table 2b do not depend on χ_r – and once a surplus opens it can over-restore while the supported population re-equilibrates downward, so a target-level stopping rule is needed for a clean $\mathcal{R}_c = 1$ (SI §S4.4). A deficit-gated flow, active when $E > B$, is the one gate that acts during the shortfall, but it retires productive land exactly when provision is short and can deepen the near-term deficit; at the tested rates conversion outpaces it, so it is necessary but insufficient and must be sized or capped. Neither gate is an unqualified good, and the gate-sign asymmetry is phase-dependent rather than a universal ranking.

3.5. NFA account composition, 1961–2023

The primary empirical window is the complete world series in NFA Edition 2026, results and metadata v1.0; 2024–2025 estimated rows are excluded. The 1961–2022 sensitivity uses the same edition and population series. Total biocapacity account value B rises from 9.7254×10^9 to 12.1428×10^9 gha (+24.86%, $\Delta \ln B = +0.221997$). Population from the workbook's UN PopStat field rises from 3.06487 to 8.09173 billion. Recomputed from these totals, B/P falls from 3.17319 to 1.50065 gha person^{-1} (−52.71%, $\Delta \ln(B/P) = -0.748840$), while the Global Ecological Footprint of Production per person rises from 2.19700 to 2.56462 gha person^{-1} (+16.73%). The production-footprint-to-biocapacity ratio E_{prod}/B rises from 0.69236 to 1.70901 and first exceeds one in 1973. The exact identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ gives $+0.970837 - 0.748840 = +0.221997$; this is an extensive/intensive accounting split, not a causal decomposition. The same-edition 1961–2022 sensitivity gives $\Delta \ln B = +0.219497$, a 52.41% fall in B/P, a 17.70% rise in E_{prod}/P, and the same first-crossing year (SI §S5).

The land-type components make the NFA account composition explicit (Fig. 3; Table 3). Cropland biocapacity contributes +24.06 percentage points to the +24.86% aggregate change, about 96.8% of the net percentage change. Forest biocapacity contributes −5.95 points, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland account sum, which includes built-up land, is nearly flat (1.0092×, or +0.80 points) only

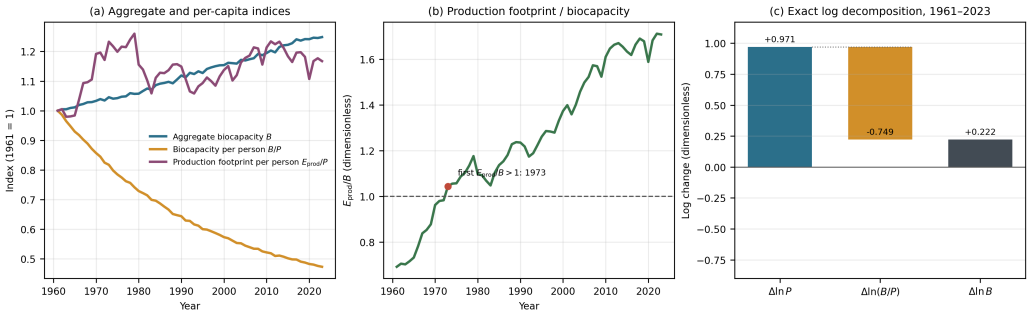


Figure 2: Aggregate and per-capita movements in NFA Edition 2026 (world, 1961–2023; complete observations). B is total biocapacity, E_{prod} is the workbook's Global Ecological Footprint of Production, and P is the single UN PopStat population series; per-capita values are recomputed from totals. (a) Indices (1961 = 1) of B, B/P, and E_{prod}/P. (b) E_{prod}/B, first above one in 1973. (c) Exact log identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$. Account values are in gha; plotted indices, ratios, and log changes are dimensionless. The 2024–2025 estimated rows are excluded.

because it nets the built-up increase against declines in other categories. This is measured composition of NFA account values. It does not establish that forest biomass, soil condition, wetland extent, fish biomass, or the model's A_c declined; built-up land is an accounting class, not ecological capital. Component values sum to total biocapacity within rounding at the checked endpoints.

The supplementary crop-proxy Divisia sensitivity is intentionally not treated as an NFA component decomposition. It uses external cropland-area and cereal-yield series with a stated 1961 account anchor; its $C = B - \hat{b}_f A_f$ residual is not an observed non-cropland account or a physical-capital measure (SI §S5).

3.6. Falsifiable predictions

The model's testable content, framed against two-book series (B, A_c, A_f and land-cover/yield proxies) rather than the aggregate, is the following.

P1. **Mask signature.** In the baseline frozen-demand run, the model predicts a finite interval with $\dot{A}_c < 0$ and $\dot{B} > 0$ (Fig. 1a). The conversion contribution is positive exactly when $\Delta b_{\text{conv}} > 0$, but the net sign of $\dot{B}$ also depends on maintenance/retirement, quality growth, debt, technology, and restoration. With baseline shares $\sigma_f = \sigma_c = 1$, active conversion requires $R_B = E/B > 1$; in the reserved-flow extension with

Table 3: NFA Edition 2026 biocapacity-account endpoint ratios and contributions. Contributions are $100\,[B_i(T) - B_i(1961)]/B(1961)$ percentage points; endpoint ratios are $B_i(T)/B_i(1961)$. The 2022 sensitivity is from the same edition. All entries are account values, not physical-stock measurements.

Component	2023 ratio	2023 contribution (pp of B_{1961})	2022 sensitivity ratio	2022 contribution (pp of B_{1961})
Cropland	2.8701	+24.0577	2.8514	+23.8183
Grazing land	0.9828	−0.2688	0.9815	−0.2890
Forest biocapacity	0.8996	−5.9525	0.9019	−5.8137
Fishing grounds	0.9627	−0.4212	0.9596	−0.4559
Built-up land†	8.7935	+7.4416	8.6298	+7.2854
Non-cropland sum†	1.0092	+0.7990	1.0083	+0.7267
Total biocapacity	1.2486	+24.8568	1.2454	+24.5450

†Built-up land is not ecological capital; the non-cropland account sum includes it. Component sums match total biocapacity within workbook rounding.

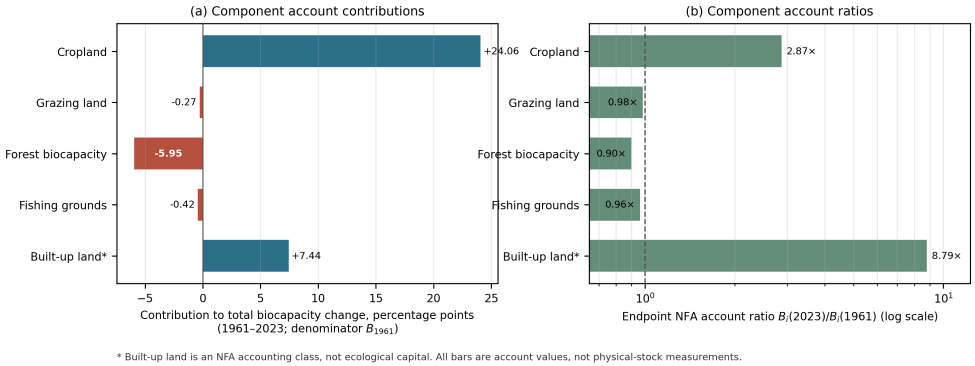


Figure 3: NFA Edition 2026 land-type biocapacity-account changes, world, 1961–2023. (a) Component contributions in percentage points of B_{1961}. (b) Endpoint ratios $B_i(2023)/B_i(1961)$. "Forest biocapacity" is the workbook's exact label. All bars are account values, not measurements of physical stocks; built-up land is an NFA accounting class, not ecological capital.

$\sigma_f, \sigma_c < 1$, conversion can occur even when $R_B \leq 1$, which is not a baseline result. A two-book series distinguishes the net composition signature (B rises while A_c falls) from a liquidation mask (E steady while A_c falls) (Proposition 1).

P2. **Window scale.** The duration of the mask is set by the slow eigenvalue λ_1 (through τ_{conv}, μ, η_f, b_c/b_f) and its amplitude by Δb_{conv} and the deficit, not by a fixed horizon; the bulk of conversion precedes the closing of the deficit by one to two e-folding times.

P3. **No critical-slowing precursor.** The return time does not grow as demand approaches the ceiling, and the drawdown depth is continuous in demand (Fig. 1c); the observable warning is the capital book approaching its typed floor (Proposition 2). This is a negative prediction: a catastrophic-shift early-warning indicator is not expected to precede the two-land drawdown.

P4. **Demand threshold, not delay threshold.** The operational initial-demand threshold for approaching the capital floor rises with initial capital (Table 2b) and is insensitive to the regeneration delay (identical to 10^{-3} for $\tau_g = 5$–$40\,\text{yr}$), because the regenerative share of capital value is small and quality is monotone; a one-stock regeneration cliff is not expected.

P5. **Recovery requires a gated flow.** Reducing demand does not restore the capital book; recovery requires an explicit restoration flow whose gate sign determines whether it is inert during the shortfall (surplus-gated; thresholds unchanged by χ_r) or deepens it (deficit-gated).

The identifiability statement of Proposition 1(iii) is not a prediction but a constraint on testing: P1–P5 can be tested only with independent A_c and b_f proxies, never with the aggregate alone.

4. Discussion

4.1. What the two-land structure changes, and what it removes

The results depart from the single-stock, single-aggregate reading of the environment that dominates the resource and sustainability literature (Schaefer, 1954; May, 1973; Meadows et al., 1972, 2004; Wackernagel and Rees,

1996; Brander and Taylor, 1998) in three respects, each of which is a negative result as much as a positive one. First, the conversion contribution can raise aggregate capacity while capital-land area A_c falls, but this bounded composition effect does not by itself determine the net change in the aggregate: Proposition 1 gives the exact sign condition for the conversion contribution, and Proposition 3 explains why the technology/debt mask cannot persist. Second, in the reduced frozen-demand case of Proposition 2 ($q = q_{\max}$, $D = 0$, technology wave off, and restoration disabled), the balance relation is strictly monotone and its floor collision is not a fold. There is no hysteresis of that equilibrium relation or critical slowing down; these are branch-level statements, not claims that realized states are independent of history. With conversion but no return flow, A_c remains below its pre-overshoot value after conversion stops, while the equilibrium itself has $\dot{A}_c = 0$ (Table 4). Third, recovery is not the reverse of degradation; it requires an explicit, gated flow, and the gate sign is a design decision with phase-dependent consequences.

The negative results have a direct bearing on early-warning practice. Generic indicators of critical slowing down (rising autocorrelation and variance) are motivated by the approach to a fold (Scheffer et al., 2001). In a two-book system in which the drawdown is a boundary collision, such indicators would not be expected to fire, and their silence should not be read as safety. The informative signal is the capital book itself, which requires the two-book series.

Several quantities that circulate for models of this kind are deliberately not claimed here. The two-gain Hopf classification of which delay destabilizes, the associated critical delays, and the reference eigenvalue of order $+0.6\,\mathrm{yr}^{-1}$ belong to a single-stock fast–slow reduction that we retain only as a comparator in SI §S7, where its equations, equilibrium family and figures are given; none of its features is carried into the two-land results. The "18–20 yr regeneration cliff" of the one-stock comparator does not survive the two-land structure (Table 4). What is structural in the two-land model is the composition logic and the identifiability limit; what is conditional on effective parameters is the sign and size of Δb_{conv} and the location of the ceiling.

4.2. Measurement implications

The methodological result of the paper is the identifiability limit. Biocapacity in global hectares is a value-weighted scalar, not a count of the underlying assets. Given only that a portfolio's value rose by 20%, one cannot tell

Table 4: Structural properties and negative results in the two-land model, each derived under the two-land structure rather than inherited from a one-stock comparator.

Phenomenon	Present?	Status in the two-land model
Capital-land catastrophic fold	No	$B_{eq}(A_c)$ is strictly monotone; the balance point is unique; the transition is a typed-floor collision at $E_{\text{ceil}} \approx 1.138\,\text{gha}$ (Proposition 2)
Critical slowing down	No	deficit-side dynamics are affine; $\lambda_1 \approx -0.054\,\text{yr}^{-1}$ is constant along the branch and does not tend to 0
Hysteresis of the reduced frozen-demand equilibrium relation	No	Under Proposition 2's restrictions, $B_{eq}(A_c)$ is strictly monotone and has at most one feasible balance point; this does not imply that realized states are history-independent
Post-overshoot path dependence of realized A_c	Yes, conditional	With $R_{rc} = 0$, conversion is one-way; after the shortfall closes, A_c remains below its pre-overshoot level, while q may continue evolving separately
Perpetual decline at a frozen-demand equilibrium	No	At a feasible balance point $u_c = 0$ and $\dot{A}_c = 0$; the lower area persists in the no-restoration trajectory but does not keep declining
Discontinuous transition with demand	No	drawdown depth is continuous and monotone in initial population P_0 (cap; Fig. 1c); thresholds are tolerance-dependent bands
Regeneration-delay cliff	No	operational initial-population floor-proximity threshold identical to three decimal places for $\tau_g = 5$–$40\,\text{yr}$; the regenerative share of capital value is small and q is monotone
Sustained boom–bust cycle	No	single overshoot pulse; q is monotone; $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold of the delayed logistic population loop (Hutchinson, 1948)
Allee-type rescue threshold in capital	No	no capital level below which recovery is impossible other than the floor itself; the lever is demand or restoration, not the lag

whether every holding grew a little or some made large gains while others quietly fell; the scalar gloss hides the composition. This is the generic limit of any value-weighted composite (Nardo et al., 2008), and Proposition 1(iii) states it for biocapacity: the aggregate identifies one linear combination of the areas, and the capital-book parameters enter only through identifiable combinations. The aggregate is therefore a weak-sustainability index in the precise sense of the companion assessment work (Abaee, 2026b): it is the linear, perfectly substitutable member of the aggregator family (its Lemma A), and it can satisfy an aggregate floor while the typed floor on A_c is violated by exactly the amount that the fast-land yield gains against the capital loss. In that family no positive-elasticity aggregator is uniformly safe and only the typed, non-compensating one is (its Theorem S2(ii)). Under exact-tube semantics, in which a transition is safe only if every state along it satisfies the constraints (Abaee, 2026b), the passage during which A_c falls is itself a violation even if the endpoint recovers – a reason to report through-time trajectories and boundary curves, not endpoint attractors.

For the National Ecological Footprint and Biocapacity Accounts, the implication is concrete but bounded. In Edition 2026, world biocapacity account value B rises 24.86% over 1961–2023, while B/P falls 52.71%; the global Ecological Footprint of Production per person rises 16.73%, and E_{prod}/B first exceeds one in 1973. At the account-component level, cropland contributes +24.06 percentage points of the aggregate change, Forest biocapacity −5.95, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland account sum is nearly flat only because it combines those movements, including the large built-up increase. The 1961–2022 same-edition sensitivity is reported separately (SI §S5); no prior-edition values are blended into this series. These are movements in NFA account values expressed in gha. They are not measurements of forest biomass or condition, soil carbon, wetland extent, fish biomass, or other physical stocks; the decline in Forest biocapacity is not evidence by itself of a measured physical-stock decline. The built-up component is an accounting class, not ecological capital. Reporting component ratios, contributions, vintage, and endpoint sensitivity alongside B exposes accounting composition without converting it into a physical-stock claim (Lo et al., 2026; Borucke et al., 2013; Blomqvist et al., 2013; Giampietro and Saltelli, 2014; Galli et al., 2016; van den Bergh and Grazi, 2015).

The model's capital decline has a different evidential status from the NFA movements. In the conditional frozen-demand baseline (Fig. 1a), model area

A_c falls from 1.500 to 0.807659 ha (-46.16%) while model aggregate B rises from 0.594 to 0.825 gha ($+39\%$). The quality state q evolves separately, from 0.9 to 1.0; because $q_0 > q^* = q_{\max}/2$, the per-hectare value $\Upsilon_c(q)$ falls from 0.05576 to 0.05000 gha ha^{-1} (-10.33%). The modeled capital contribution $\Upsilon_c(q)A_c$ therefore falls from 0.08364 to 0.04038 gha (-51.72%). With restoration disabled, one-way conversion leaves A_c below its pre-overshoot level after the shortfall closes; at the balance point conversion stops and $\dot{A}_c = 0$, so the lower area persists rather than declining perpetually. These values are conditional outputs of a representative model run, not observations or estimates of a global physical-stock trend.

The paper does not populate a cross-asset physical-stock demonstrator. A defensible global inventory over the NFA period would need common spatial and temporal boundaries, compatible definitions, uncertainty, and component-specific observations of extent, condition, or stock; the sources assembled here do not provide that matched inventory. The SI physical-stock scoreboard is therefore an unpopulated specification, not data or an empirical result. Any physical assessment should preserve a nonadditive vector: report extent (for example, forest or wetland area) separately from condition or stocks (for example, biomass, soil organic carbon for a stated depth, fish spawning-stock biomass, and freshwater storage), and keep annual flows and pressures separate from stocks. Declare overlap so nested measures are not double-counted. The components can be interpreted jointly without scalarization; a single index requires a declared valuation basis, normalization, weights, and permitted substitution. This follows the separation of ecosystem extent, condition, services, and monetary presentations in SEEA Ecosystem Accounting (United Nations et al., 2021). No such physical-stock composite is constructed in this study.

The supplementary crop proxy illustrates the limits of an observation operator rather than overcoming them. It combines FAOSTAT-derived cropland area and cereal yield with an explicit 1961 NFA-cropland-account anchor. Because the cereal series is a yield sentinel and the physical-area and NFA-account boundaries are not identical by construction, $C = B - \hat{b}_f A_f$ remains an accounting residual: it can include calibration and boundary mismatch, other account components, and conversion-factor changes. It is not the model contribution $\Upsilon_c(q)A_c$ and cannot be read as physical capital. Such an interpretation would require independently measured, boundary-consistent area, condition, yield, and regeneration inputs plus a validated mapping and valuation. Forest area may serve as one sentinel for part of

the capital class, but it is not a complete measure of the model state or of heterogeneous ecological assets.

4.3. Policy operators

Three operators follow from the typed-floor structure and are developed in SI §S6. Freezing conversion – capping A_f or flooring A_c – is the single operator that guarantees $A_c \geq A_c^{\min}$; a governor that observes only the composite B cannot guarantee a capital-land floor, and the failure is informational rather than dynamical. Restoration is protective only when surplus-gated, and a deficit-gated flow can deepen the shortfall it is meant to relieve; this is the protective, as opposed to mobilizing, institutional signature of the delay companion (Abaee, 2026a). A Half-Earth reservation (Wilson, 2016), $E \leq \sigma B$ with $\sigma < 1$, succeeds without liquidation on the nominal path, but this is a nominal-path consequence until it is certified against a declared disturbance class under a preregistered retention rule, and we state it as such.

4.4. Relation to companion work

The accounting layer – the closed typed-flux ledger, conservation and donor-limitation rules, depletion arithmetic and the no-non-negative-weighting aggregation obstruction – is the object of the ledger companion (Abaee, 2026c), which owns the static layer, cedes delay dynamics to its companions and registers a two-pool dynamical model as an open gap; the present paper supplies that minimal dynamical realization for the global-hectare case. The aggregation theorem that places B at the weak extreme of the aggregator family is from the assessment companion (Abaee, 2026b), and the institutional-delay regime is from the governance companion (Abaee, 2026a); the delays here are ecological (regeneration, demography), not institutional. We cite, and do not re-derive, those results.

4.5. Limitations

The capital land is an effective class, not a single land type; the single regeneration timescale, the composition diagnostic and the ceiling are effective-parameter statements, and the full vector decomposition is cited rather than re-derived. Most parameters are representative rather than estimated; b_c, $b_{G,c}$ and ρ_c are not identifiable from a cross-section, and the regeneration anchors bound a band rather than calibrate ρ_c (the anchors concern the regeneration rate, not the lag τ_g). The per-capita footprint is held constant to isolate the

stock–flow–demand feedbacks; endogenizing e is an extension. The composition window is bounded and demand-driven; without a sustained shortfall or a technology/demand feedback it closes at the balance point. Stability at $S = 0$ is one-sided because the positive part is exact. Drawdown thresholds are tolerance-dependent bands; a small margin next to that band is reported as such and not as a sharp boundary. The debt-repayment rate is set physical and non-singular ($\eta = 0.05\,\mathrm{yr}^{-1}$); the $\eta \to 0$ case removes the debt equilibrium and is excluded by a floor, and outcomes are insensitive to η over a modest positive range. The accounts do not separate yield from area, which is the identifiability limitation the paper is about rather than an estimation error. Finally, the delayed-regeneration term is implemented at the current quality in the deposited solver (Section 2.4); we have verified that applying the delay changes no reported quantity by more than 1%, and the corrected variant is deposited alongside.

5. Conclusions

Aggregate biocapacity alone does not identify its composition. In the two-land model, conversion contributes positively to B when fast-land yield exceeds the effective per-hectare value of capital land, while A_c falls; the net change in B depends on the other terms (Proposition 1). Under the stated reduced frozen-demand assumptions, the equilibrium relation is single-valued and has no fold, hysteresis, or critical slowing down. That result is distinct from path dependence of the realized state: with restoration disabled, one-way conversion leaves A_c below its pre-overshoot value after the shortfall closes, but the equilibrium itself has $\dot{A}_c = 0$. In the representative run, A_c falls 46.16%, $\Upsilon_c(q)$ falls 10.33%, and $\Upsilon_c(q)A_c$ falls 51.72% while model B rises 39%; q evolves separately, and none of these are empirical global-stock estimates.

The Edition 2026 NFA world accounts provide a different result: from 1961 to 2023, total B rises 24.86%, B/P falls 52.71%, and the global production footprint per person rises 16.73%. Cropland contributes +24.06 percentage points to the total biocapacity-account change; Forest biocapacity contributes −5.95 points, while built-up land contributes +7.44 points as an accounting class, not ecological capital. These are account-composition signals, not measurements of biomass, soil condition, wetland extent, fish biomass, or A_c. The 1961–2022 sensitivity uses the same NFA edition and population series. No comparable global physical-stock inventory was as-

sembled, so the scoreboard is an unpopulated specification. Heterogeneous physical-stock measures remain a nonadditive vector unless common spatial and temporal boundaries, normalization or valuation, and substitution rules are declared. The supplementary $C = B - \hat{b}_f A_f$ is only a proxy residual; it is not physical capital or the model's $\Upsilon_c(q)A_c$. Report NFA components alongside their aggregate, and do not translate account movements into physical-stock decline without independent, boundary-consistent observations and a validated mapping.

Acknowledgments

The author thanks Global Footprint Network for making its data publicly available; Mathis Wackernagel for his thoughtful review; the editors of *Ecological Modelling* for their trust and support; and Arena.ai for its ingenuity and online tools.

Declarations

Data availability. The published v44 source and reproducibility package, which is the predecessor rather than the source of the present empirical revision, remains available at https://github.com/MIKEAA2020/general-sustainability/tree/main/agent%202%20productivity%20illusion/manuscript_ECOMOD_v44_src. The v45 working package includes the Edition 2026 NFA workbook, the independently sourced OWID/FAOSTAT cropland-area and cereal-yield inputs, derived extracts, analysis scripts, regenerated figures, and manuscript/SI sources. Its public archive path will be assigned only after the version, citation, and build checks are complete; this working revision has not been published. The NFA workbook and release notes are available from York University's Ecological Footprint Initiative data page (https://footprint.info.yorku.ca/data/); source-series pages are cited in the references. No comparable global physical-stock inventory was assembled.

Funding. No funding was received.

Competing interests. The author declares no competing interests.

CRediT authorship contribution statement. **Amin Abaee:** Conceptualization; Methodology; Software; Formal analysis; Writing – original draft; Writing – review & editing.

Declaration of generative AI in scientific writing. DeepSeek AI and Claude (Anthropic) assisted with computational-reproducibility checks under the author's supervision; the author reviewed and edited the content and takes full responsibility for the publication.

References

Abaee, A. (2026a). *Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter* (v41 manuscript, 22 September 2026). GitHub, commit c1168a898f43348424ae9ea887a79c0def4d6747: https://github.com/MIKEAA2020/general-sustainability/blob/c1168a898f43348424ae9ea887a79c0def4d6747/arena%20agent%201/paper%20rewrites/latex/paper4_delay_dynamics_v41.pdf. The earlier Zenodo deposit (record 22554217, v30; DOI: https://doi.org/10.5281/zenodo.22554217) is archived under the title *Delay-Induced Regime Change in Harvested Stocks: The Mobilising and Protective Channels of Institutional Feedback, and the Review Interval as Control.*

Abaee, A. (2026b). *Aggregate Indices and Transition Safety: A Quantifier-Order Separation Between Scalarized and Coordinate-Wise Feasibility* (v59 manuscript, 22 September 2026). GitHub, commit 18344877c6b7453e9a60234b60a85fb521d5a998: https://github.com/MIKEAA2020/general-sustainability/blob/18344877c6b7453e9a60234b60a85fb521d5a998/arena%20agent%201/paper%20rewrites/latex/paper1_assessment_separation_v59.pdf. The earlier Zenodo deposit (record 22545740, v22; DOI: https://doi.org/10.5281/zenodo.22545740) is archived under the title *The Limits of Compensatory Aggregation: A Formal Separation of Weak and Strong Sustainability Assessment.*

Abaee, A. (2026c). *Typed Flux Ledgers and Depletion Arithmetic: Conservation, Componentwise Diagnostics, and the Semantics of Depletion Horizons* (v32 manuscript). GitHub, commit 426dadb48522d50f1033ff503717192ef168f1fa: https://github.com/MIKEAA2020/general-sustainability/blob/426dadb48522d50f1033ff503717192ef168f1fa/arena%20agent%201/paper%20rewrites/latex/paper3_material_ledgers_v32.pdf.

The related Zenodo deposit (record 22554177; DOI: https://doi.org/10.5281/zenodo.22554177) preserves an earlier v31 PDF under the same title.

Anderies, J. M. (2000). On modeling human behavior and institutions in simple ecological economic systems. *Ecological Economics*, 35(3), 393–412.

Becker, W., Saisana, M., Paruolo, P. & Vandecasteele, I. (2017). Weights and importance in composite indicators: Closing the gap. *Ecological Indicators*, 80, 12–22.

Blomqvist, L., Brook, B. W., Ellis, E. C., Kareiva, P. M., Nordhaus, T. & Shellenberger, M. (2013). Does the shoe fit? Real versus imagined ecological footprints. *PLoS Biology*, 11(11), e1001700.

Borucke, M., Moore, D., Cranston, G., Gracey, K., Iha, K., Larson, J., Lazarus, E., Morales, J. C., Wackernagel, M. & Galli, A. (2013). Accounting for demand and supply of the biosphere's regenerative capacity: The National Footprint Accounts' underlying methodology and framework. *Ecological Indicators*, 24, 518–533.

Brander, J. A. & Taylor, M. S. (1998). The simple economics of Easter Island: A Ricardo–Malthus model of renewable resource use. *American Economic Review*, 88(1), 119–138.

Calvin, K. & Bond-Lamberty, B. (2018). Integrated human–Earth system modeling — state of the science and future directions. *Environmental Research Letters*, 13(6), 063006.

Cohen, J. E. (1995). Population growth and Earth's human carrying capacity. *Science*, 269(5222), 341–346.

Dasgupta, P. (2021). *The Economics of Biodiversity: The Dasgupta Review.* London: HM Treasury.

Diamond, J. (2005). *Collapse: How Societies Choose to Fail or Succeed.* New York: Viking.

Donges, J. F., Winkelmann, R., Cornell, S. E., Lucht, W., Dyke, J. G., Rockström, J., Heitzig, J. & Schellnhuber, H. J. (2017). Closing the loop: Reconnecting human dynamics to Earth system science. *The Anthropocene Review*, 4(2), 151–157.

Erb, K.-H., Luyssaert, S., Meyfroidt, P., Pongratz, J., Don, A., Kloster, S., Kuemmerle, T., Fetzel, T., Fuchs, R., Hasegawa, T., Haberl, H., et al. (2017). Land management: data availability and process understanding for global change studies. *Global Change Biology*, 23(2), 512–533.

Fischer, S. M., Joy, M. K., Abrahamse, W., Milfont, T. L. & Petherick, L. M. (2022). The use and misuse of composite environmental indices. *bioRxiv*, 2022.03.15.484501. https://doi.org/10.1101/2022.03.15.484501.

Global Footprint Network (2009). *Ecological Footprint Standards 2009*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/Ecological_Footprint_Standards_2009.pdf.

Global Footprint Network (2019). *Working Guidebook to the National Footprint Accounts 2019*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/National_Footprint_Accounts_Guidebook_2019.pdf.

Galli, A., Giampietro, M., Goldfinger, S., Lazarus, E., Lin, D., Saltelli, A., Wackernagel, M. & Müller, F. (2016). Questioning the ecological footprint. *Ecological Indicators*, 69, 224–232.

Giampietro, M. & Saltelli, A. (2014). Footprints to nowhere. *Ecological Indicators*, 46, 610–621.

Haberl, H., Erb, K.-H., Krausmann, F., Gaube, V., Bondeau, A., Plutzar, C., Gingrich, S., Lucht, W. & Fischer-Kowalski, M. (2007). Quantifying and mapping the human appropriation of net primary production in earth's terrestrial ecosystems. *PNAS*, 104(31), 12942–12947.

Henderson, K. & Loreau, M. (2018). How ecological feedbacks between human population and land cover influence sustainability. *PLoS Computational Biology*, 14(8), e1006389.

Hutchings, J. A. & Reynolds, J. D. (2004). Marine fish population collapses: consequences for recovery and extinction risk. *BioScience*, 54(4), 297–309.

Hutchinson, G. E. (1948). Circular causal systems in ecology. *Annals of the New York Academy of Sciences*, 50(4), 221–246.

Krausmann, F., Erb, K.-H., Gingrich, S., Haberl, H., Bondeau, A., Gaube, V., Lauk, C., Plutzar, C. & Searchinger, T. D. (2013). Global human appropriation of net primary production doubled in the 20th century. *PNAS*, 110(25), 10324–10329.

Kuang, Y. (1993). *Delay Differential Equations, with Applications in Population Dynamics.* Academic Press.

Lin, D., Hanscom, L., Murthy, A., Galli, A., Evans, M., Neill, E., et al. (2018). Ecological footprint accounting for countries: updates and results of the National Footprint Accounts. *Sustainability*, 10(12).

Lo, K., Basnet, N., Miller, E., Srikanthalingam, B., Foley, B., Van Berkum, J. L., Long, A. H., Toneva, P., Ermina, M. & Klinkenberg, C. (2026). *National Ecological Footprint and Biocapacity Accounts, Edition 2026 (1961–2025): Results and metadata, version 1.0.* Prepared for Footprint Data Foundation by researchers at York University and the University of Iceland. Workbook: https://footprint.info.yorku.ca/files/2026/04/NEFBA_Data_global_national_ecological_footprint_biocapacity.xlsx; release notes: https://footprint.info.yorku.ca/files/2026/04/NEFBA_ReleaseNotes.pdf; data page: https://footprint.info.yorku.ca/data/.

May, R. M. (1973). *Stability and Complexity in Model Ecosystems.* Princeton: Princeton University Press.

Meadows, D. H., Meadows, D. L., Randers, J. & Behrens, W. W. (1972). *The Limits to Growth.* New York: Universe Books.

Meadows, D. H., Randers, J. & Meadows, D. L. (2004). *The Limits to Growth: The 30-Year Update.* White River Junction, VT: Chelsea Green.

Meadows, D. L., Behrens, W. W., Meadows, D. H., Naill, R. F., Randers, J. & Zahn, E. (1974). *Dynamics of Growth in a Finite World.* Cambridge, MA: Wright-Allen Press.

Meyer, P. S. (1994). Bi-logistic growth. *Technological Forecasting and Social Change*, 47(1), 89–102.

Meyer, P. S. & Ausubel, J. H. (1999). Carrying capacity: A model with logistically varying limits. *Technological Forecasting and Social Change*, 61(3), 209–214.

Motesharrei, S., Rivas, J., Kalnay, E., Asrar, G. R., Busalacchi, A. J., Cahalan, R. F., Cane, M. A., Colwell, R. R., Feng, K., Franklin, R. S., et al. (2016). Modeling sustainability: Population, inequality, consumption, and bidirectional coupling of the Earth and human systems. *National Science Review*, 3(4), 470–494.

Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffmann, A. & Giovannini, E. (2008). *Handbook on Constructing Composite Indicators: Methodology and User Guide.* Paris: OECD Publishing.

Neubauer, P., Jensen, O. P., Hutchings, J. A. & Baum, J. K. (2013). Resilience and recovery of overexploited marine populations. *Science*, 340(6130), 347–349.

Our World in Data (2026a). *Cropland area.* FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cropland-area.

Our World in Data (2026b). *Cereal yield.* FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cereal-yield.

Pearl, R. (1925). *The Biology of Population Growth.* New York: Alfred A. Knopf.

Poeplau, C., et al. (2011). Temporal dynamics of soil organic carbon after land-use change. *Global Change Biology*, 17(7), 2415–2427.

Poorter, L., et al. (2016). Biomass resilience of Neotropical secondary forests. *Nature*, 530(7590), 211–214.

Schaefer, M. B. (1954). Some aspects of the dynamics of populations important to the management of the commercial marine fisheries. *Bulletin of the Inter-American Tropical Tuna Commission*, 1(2), 25–56.

Scheffer, M., Carpenter, S., Foley, J. A., Folke, C. & Walker, B. (2001). Catastrophic shifts in ecosystems. *Nature*, 413(6856), 591–596.

Simon, J. L. (1996). *The Ultimate Resource 2.* Princeton: Princeton University Press.

Tainter, J. A. (1988). *The Collapse of Complex Societies.* Cambridge: Cambridge University Press.

United Nations et al. (2021). *System of Environmental-Economic Accounting—Ecosystem Accounting (SEEA EA).* White cover publication, pre-edited text subject to official editing. New York: United Nations. https://seea.un.org/ecosystem-accounting.

United Nations (2022). *World Population Prospects 2022: Summary of Results.* New York: Department of Economic and Social Affairs, Population Division.

van den Bergh, J. C. J. M. & Grazi, F. (2015). Reply to the first systematic response by the Global Footprint Network to criticism: a real debate finally? *Ecological Indicators*, 58, 458–463.

van den Bergh, J. C. J. M. & Rietveld, P. (2004). Reconsidering the limits to world population: Meta-analysis and meta-prediction. *BioScience*, 54(3), 195–204.

Wackernagel, M. & Rees, W. (1996). *Our Ecological Footprint: Reducing Human Impact on the Earth.* New Society Publishers.

Wilson, E. O. (2016). *Half-Earth: Our Planet's Fight for Life.* New York: Liveright.